

Practical 1

Aim:Linux System Administration and File Management for Marketing Analytics Team

Problem definition:Scenario: You have recently joined DataAnalytics Inc. as a Linux System Administrator. The company runs its analytics platform on Linux servers where project data, reports, scripts, and user accounts must be managed efficiently. Your manager has assigned you the responsibility of setting up the project environment, managing files and directories, monitoring system resources, securing data, and creating backups for the Marketing Analytics team.

Objectives:

- To identify the current user and working directory in the Linux environment.
- To display basic system information, current date, and system uptime.
- To create and manage project directories and files for the Marketing Analytics team.
- To add, view, search, copy, rename, move, and delete files using Linux commands.
- To organize project data and maintain backups efficiently.
- To understand Linux file management and text processing commands.
- To monitor system resources and perform basic Linux administration tasks.

Output

1. Identify the current user.

```
krina@kri:~$ whoami  
krina
```

2. Find the current working directory.

```
krina@kri:~$ pwd  
/home/krina
```

3. Display system information.

```
krina@kri:~$ uname -a  
Linux kri 7.0.0-28-generic #28-Ubuntu SMP PREEMPT_DYNAMIC Sun Jun 21 00:50:10 UTC 2026 aarch64 GNU/Linux
```

4. Check system uptime and date.

```
krina@kri:~$ uptime  
15:29:44 up 15 min, 1 user, load average: 0.02, 0.08, 0.06
```

5. Create a directory named MarketingAnalytics.

```
krina@kri:~$ mkdir MarketingAnalytics
krina@kri:~$ cd MarketingAnalytics
krina@kri:~/MarketingAnalytics$ pwd
/home/krina/MarketingAnalytics
```

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Create subdirectories: RawData, ProcessedData, Reports.

```
krina@kri:~/MarketingAnalytics$ mkdir RawData ProcessedData Reports
krina@kri:~/MarketingAnalytics$ ls
ProcessedData  RawData  Reports
```

7. Create data and report files.

```
krina@kri:~/MarketingAnalytics$ touch RawData/customer_data.txt
krina@kri:~/MarketingAnalytics$ touch Reports/marketing_report.txt
krina@kri:~/MarketingAnalytics$ touch ProcessedData/processed_data.txt
krina@kri:~/MarketingAnalytics$ find .
.
./RawData
./RawData/customer_data.txt
./ProcessedData
./ProcessedData/processed_data.txt
./Reports
./Reports/marketing_report.txt
```

8. Add sample content.

```
krina@kri:~/MarketingAnalytics$ cat > RawData/customer_data.txt
CustomerID,Name,City
101,Raj,Ahmedabad
102,Priya,Vadodara
103,Amit,Surat
krina@kri:~/MarketingAnalytics$ cat > Reports/marketing_report.txt
Marketing Analytics Report

Campaign Performance:
Customer data collected successfully.

Prepared by Marketing Team
krina@kri:~/MarketingAnalytics$ cat > ProcessedData/processed_data.txt
Processed Customer Records
Total Customers: 3
```

9. Display file contents.

```
krina@kri:~/MarketingAnalytics$ cat RawData/customer_data.txt
CustomerID,Name,City
101,Raj,Ahmedabad
102,Priya,Vadodara
103,Amit,Surat
krina@kri:~/MarketingAnalytics$ cat Reports/marketing_report.txt
Marketing Analytics Report

Campaign Performance:
Customer data collected successfully.

Prepared by Marketing Team
```

10. List all files and directories.

```
krina@kri:~/MarketingAnalytics$ ls
ProcessedData  RawData  Reports
krina@kri:~/MarketingAnalytics$ ls -l
total 12
drwxrwxr-x 2 krina krina 4096 Aug  5 15:31 ProcessedData
drwxrwxr-x 2 krina krina 4096 Aug  5 15:30 RawData
drwxrwxr-x 2 krina krina 4096 Aug  5 15:31 Reports
krina@kri:~/MarketingAnalytics$ ls -R
.:
ProcessedData  RawData  Reports

./ProcessedData:
processed_data.txt

./RawData:
customer_data.txt

./Reports:
marketing_report.txt
```

11. Search for report files.

```
krina@kri:~/MarketingAnalytics$ find . -name "*.txt"
./RawData/customer_data.txt
./ProcessedData/processed_data.txt
./Reports/marketing_report.txt
krina@kri:~/MarketingAnalytics$ find . -name "*report*"
./Reports/marketing_report.txt
```

12. Search specific content inside files.

```
krina@kri:~/MarketingAnalytics$ grep "Priya" RawData/customer_data.txt
102,Priya,Vadodara
krina@kri:~/MarketingAnalytics$ grep "Marketing" Reports/marketing_report.txt
Marketing Analytics Report
Prepared by Marketing Team
```

13. Copy report files to Backup directory.

```
krina@kri:~/MarketingAnalytics$ mkdir Backup
krina@kri:~/MarketingAnalytics$ cp Reports/marketing_report.txt Backup/
krina@kri:~/MarketingAnalytics$ ls Backup
marketing_report.txt
```

14. Rename files.

```
krina@kri:~/MarketingAnalytics$ mv ProcessedData/processed_data.txt Reports/
krina@kri:~/MarketingAnalytics$ ls Reports
marketing_report.txt  processed_data.txt
```

15. Move files between directories.

```
krina@kri:~/MarketingAnalytics$ mv ProcessedData/processed_data.txt Reports/
krina@kri:~/MarketingAnalytics$ ls Reports
marketing_report.txt  processed_data.txt
```

16. Delete a file.

```
krina@kri:~/MarketingAnalytics$ rm Reports/processed_data.txt
krina@kri:~/MarketingAnalytics$ ls Reports
marketing_report.txt
```

17. Delete an empty directory.

```
krina@kri:~/MarketingAnalytics$ ls ProcessedData
krina@kri:~/MarketingAnalytics$ rmdir ProcessedData
krina@kri:~/MarketingAnalytics$ ls
Backup  RawData  Reports
```

Result :

The practical was completed successfully using Linux terminal commands. Basic Linux system administration and file management operations were performed, including identifying the current user, checking system information, creating directories and files, adding and viewing data, searching files and content, copying, renaming, moving, and deleting files. A backup directory was created, and the project structure was managed successfully. This practical helped in understanding how Linux commands are used to organize project data and manage files efficiently in a real-world data analytics environment.

Key Questions /Analysis:

Q1. What is the difference between absolute and relative paths?

Answer:

An **absolute path** gives the complete location of a file or folder starting from the root directory (/). It works from any location in the system.

A **relative path** gives the location of a file based on the current working directory. It is shorter but only works if you are in the correct directory.

Example:

Absolute Path: /home/krina/MarketingAnalytics/RawData/customer_data.txt

Relative Path: RawData/customer_data.txt

Q2. Why are Linux permissions important for file security?

Answer:

Linux permissions control who can read, write, or execute a file. They help protect important files from unauthorized access or accidental changes. Permissions improve data security and ensure that only authorized users can modify project files.

Q3. What is the purpose of the **grep command?**

Answer:

The **grep** command is used to search for a specific word or text inside a file. It quickly finds the required information without opening the entire file. It is very useful for searching data, reports, and log files.

Q4. How does Linux support efficient file management?

Answer:

Linux provides many commands such as `mkdir`, `ls`, `cp`, `mv`, `rm`, `find`, and `grep` to create, organize, search, copy, move, and delete files easily. It also supports file permissions and automation through shell scripts, making file management fast and efficient.

Q5. Why are backups important in enterprise environments?

Answer:

Backups create a copy of important files so they can be restored if the original data is lost, deleted, or damaged. They protect business data from system failures, accidental deletion, and cyberattacks. Regular backups help organizations avoid data loss and continue their work without interruption.

Supplementary Problems:

1. Create a shell script to automate project directory creation.

```
krina@kri:~/MarketingAnalytics$ nano create_project.sh
krina@kri:~/MarketingAnalytics$ chmod +x create_project.sh
krina@kri:~/MarketingAnalytics$ ./create_project.sh
Project Created Successfully
krina@kri:~/MarketingAnalytics$ ls MarketingProject
Backup  ProcessedData  RawData  Reports
```

2. Find the top 10 largest files in the system.

```
3.9G    /swap.img
2.9G    /run/media/krina/Ubuntu 26.04 arm64/casper/minimal.squashfs
553M    /var/lib/snapd/seed/snaps/gnome-46-2404_154.snap
493M    /run/media/krina/Ubuntu 26.04 arm64/casper/minimal.standard.squashfs
266M    /var/lib/snapd/seed/snaps/firefox_8105.snap
175M    /var/lib/snapd/seed/snaps/mesa-2404_1166.snap
159M    /snap/firefox/8105/usr/lib/firefox/libxul.so
141M    /run/media/krina/Ubuntu 26.04 arm64/casper/initrd
139M    /run/media/krina/Ubuntu 26.04 arm64/casper/minimal.standard.live.squashfs
133M    /snap/mesa-2404/1166/usr/lib/aarch64-linux-gnu/libLLVM.so.20.1
krina@kri:~/MarketingAnalytics$ 0 18 * * * cp -r ~/MarketingAnalytics ~/MarketingBackup
0: command not found
```

3. Schedule a backup task using cron.

```
krina@kri:~/MarketingAnalytics$ crontab -e
no crontab for krina - using an empty one
Select an editor. To change later, run select-editor again.
 1. /bin/nano          <---- easiest
 2. /usr/bin/vim.tiny
 3. /bin/ed

Choose 1-3 [1]: 0 18 * * * cp -r ~/MarketingAnalytics ~/MarketingBackup
crontab: installing new crontab
krina@kri:~/MarketingAnalytics$ crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow   command
```

4. Monitor CPU utilization continuously using top/htop.

```
krina@kri:~/MarketingAnalytics$ top

top - 15:56:41 up 42 min, 1 user, load average: 0.48, 0.15, 0.06
Tasks: 198 total, 1 running, 197 sleeping, 0 stopped, 0 zombie
%Cpu(s): 1.0 us, 0.2 sy, 0.0 ni, 98.7 id, 0.2 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3381.1 total, 387.3 free, 2212.1 used, 1021.1 buff/cache
MiB Swap: 3893.0 total, 3822.7 free, 70.2 used, 1169.0 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
7271	krina	20	0	1755360	412396	129064	S	1.0	11.9	0:14.62	ptxis
2609	krina	20	0	4053504	401628	109084	S	0.7	11.6	0:36.99	gnome-shell
8241	krina	20	0	14132	5304	3232	R	0.7	0.2	0:00.04	top
1601	rtkit	21	1	22620	2660	2448	S	0.3	0.1	0:00.04	rtkit-daemon
7278	krina	20	0	242344	12888	6116	S	0.3	0.4	0:00.33	ptxis-agent
7669	root	20	0	0	0	0	I	0.3	0.0	0:00.08	kworker/R:0-events
1	root	20	0	28168	17860	10484	S	0.0	0.5	0:01.28	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_workqueue_release
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_gp
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-sync_wq
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-kvfree_rcu_reclaim
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-slub_flushwq
8	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-netns
10	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/0:0H-kblockd
13	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-mm_percpu_wq
14	root	20	0	0	0	0	S	0.0	0.0	0:00.09	ksoftirqd/0
15	root	20	0	0	0	0	I	0.0	0.0	0:00.19	rcu_preempt
16	root	20	0	0	0	0	S	0.0	0.0	0:00.00	rcu_exp_par_gp_kthread_worker/0
17	root	20	0	0	0	0	S	0.0	0.0	0:00.00	rcu_exp_gp_kthread_worker
18	root	rt	0	0	0	0	S	0.0	0.0	0:00.02	migration/0
19	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/0
20	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/0
21	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/1
22	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/1
23	root	rt	0	0	0	0	S	0.0	0.0	0:00.02	migration/1
24	root	20	0	0	0	0	S	0.0	0.0	0:00.07	ksoftirqd/1
26	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/1:0H-kblockd
27	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kdevtmpfs
28	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-inet_frag_wq
29	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_kthread
30	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_rude_kthread
31	root	20	0	0	0	0	S	0.0	0.0	0:00.00	auditd
32	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kblockd

5. Generate a disk usage report for all project directories.

```
krina@kri:~/MarketingAnalytics$ du -sh *
8.0K    Backup
4.0K    create_project.sh
20K     MarketingProject
8.0K    RawData
8.0K    Reports
krina@kri:~/MarketingAnalytics$ du -h .
8.0K    ./Backup
8.0K    ./RawData
4.0K    ./MarketingProject/Backup
4.0K    ./MarketingProject/RawData
4.0K    ./MarketingProject/ProcessedData
4.0K    ./MarketingProject/Reports
20K     ./MarketingProject
8.0K    ./Reports
52K     .
```

Tools /Technology used:

- Ubuntu Linux
- Linux Terminal
- Bash Shell
- Basic Linux Commands (whoami, pwd, mkdir, touch, ls, cat, find, grep, cp, mv, rm, rmdir, du, top, crontab)

Conclusion

In this practical, I learned how to perform basic Linux system administration and file management tasks using the Linux terminal. I created and managed directories and files, searched data, created backups, and monitored system resources using different Linux commands. This practical improved my understanding of Linux command-line operations and showed how Linux is used to manage files and data efficiently in Big Data Analytics environments.